A rational counterexample to the nonreducibility characterization

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Task 1: the source question. Conjecture 5.3 of Kiss and Somlai [1] concerns associative, idempotent, strictly coordinatewise monotone operations on a totally ordered Abelian group $(X, +)$. It proposes that a nonreducible operation must have odd arity and be conjugate, by a monotone bijection $X \to X$, to the alternating sum formed using the given group addition. Here reducibility means iteration of an associative binary operation on the same set; no monotonicity of that binary operation is assumed. The printed sum starts with a minus sign. We refute the substantive interpretation starting with a plus sign, as in the source's Example 4.10. Thus the counterexample does not depend on that typographical issue.

Theorem 1. *There is an associative, idempotent operation $F : \mathbb{Q}^3 \to \mathbb{Q}$, strictly increasing in its first and third variables and strictly decreasing in its second variable, which is nonreducible and for which no bijection $g : \mathbb{Q} \to \mathbb{Q}$ satisfies*

$$F(x, y, z) = g^{-1}\big(g(x) - g(y) + g(z)\big). \tag{1}$$

In particular, the characterization in Conjecture 5.3 fails for the usual ordered Abelian group $(\mathbb{Q}, +)$, even after correction of the sign.

Lemma 1. *Let $H = \mathbb{Z} + \sqrt{2}\mathbb{Z} \subset \mathbb{R}$ with its usual order. There exists an increasing bijection $h : \mathbb{Q} \to H$ with $h(0) = 0$ and $h(1) = 1$. However, $1/2 \notin H$.*

Proof. The set H is a countable additive subgroup, unbounded in both directions. To see that it is dense, divide $[0, 1)$ into N equal half-open intervals and consider the $N + 1$ fractional parts of $j\sqrt{2}$, $0 \le j \le N$. Two lie in the same interval; their nonzero difference, with sign chosen positive, gives $d \in H$ with $0 < d < 1/N$. For any real $a < b$, choose such a $d < b - a$. Then $(\lfloor a/d \rfloor + 1)d$ lies in $(a, b) \cap H$.

For completeness, construct h by back-and-forth. Enumerate both countable sets and start with $0 \mapsto 0$, $1 \mapsto 1$. At alternate steps, take the first rational not yet in the finite domain and map it to an unused element of H between the images of its nearest domain neighbors; when a neighbor is missing use the corresponding unbounded interval. At the other steps, take the first element of H not yet in the range and choose an unused rational in the corresponding interval. Density and absence of endpoints make each choice possible. The union is increasing, defined on every rational, and onto H.

Finally, $a + b\sqrt{2} = 1/2$ with $a, b \in \mathbb{Z}$ would force $b = 0$ by irrationality of $\sqrt{2}$, and then $a = 1/2$, a contradiction. □

Proof of the theorem. Fix h from Lemma 1 and define

$$F(x, y, z) = h^{-1}\big(h(x) - h(y) + h(z)\big). \tag{2}$$

Closure of H under addition and subtraction makes this a total operation on $\mathbb{Q}$. Its strict monotonicity in each coordinate and its idempotency follow immediately from the increasing bijection h. Writing $T(u, v, w) = u - v + w$ on H, all three ternary associativity expressions

$$T(T(u_1, u_2, u_3), u_4, u_5), \quad T(u_1, T(u_2, u_3, u_4), u_5), \quad T(u_1, u_2, T(u_3, u_4, u_5))$$

equal $u_1 - u_2 + u_3 - u_4 + u_5$. Transport through h proves associativity of F in the source's precise sense.

Suppose an associative binary operation on $\mathbb{Q}$ reduced F. Transporting it through h would give an associative operation $*$ on H such that

$$(u * v) * w = u * (v * w) = u - v + w.$$

Let $e = 0 * 0$. For every $z \in H$,

$$e * z = (0 * 0) * z = T(0, 0, z) = z, \qquad z * e = z * (0 * 0) = T(z, 0, 0) = z.$$

Thus e is a two-sided identity. Consequently $T(e, x, e) = (e * x) * e = x$ for all $x \in H$. But the definition of T gives $T(e, x, e) = 2e - x$. Hence $2(x - e) = 0$ for every $x \in H$, contradicting, for instance, $x = e + 1$. This proves nonreducibility without any assumptions on a proposed binary reduction except associativity.

On the other hand, equation $F(x, 0, x) = 1$ has no solution: by (2) and the normalization of h, it is equivalent to $2h(x) = 1$, impossible by Lemma 1. If a bijection $g : \mathbb{Q} \to \mathbb{Q}$ satisfied (1), the rational

$$x = g^{-1}\left(\frac{g(0) + g(1)}{2}\right)$$

would solve precisely that equation. This contradiction excludes every bijection g, and therefore every monotone one. □

Scope. The group addition in the conjecture is fixed. Formula (2) transports addition from the different ordered group H, which has the same order type as $\mathbb{Q}$ but different divisibility properties. The theorem disproves the universally quantified characterization at $n = 3$. It makes no claim about the restricted case $X = \mathbb{R}$, or about a formulation allowing a different target group for the conjugating map. The original printed minus-first sum cannot represent an idempotent operation on $\mathbb{Q}$ either: on the diagonal it would require $-g(x) = g(x)$ for every x.

Proposed follow-up tasks (not claimed solved here).

2. Extend the counterexample to every odd arity $n \geq 3$, proving all associativity identities and nonreducibility without invoking the source's reducibility theorems.

3. Classify when the ternary operations $u - v + w$ on two dense additive subgroups of $\mathbb{R}$ are conjugate by an increasing bijection; determine whether a translation followed by positive scaling is necessary and sufficient.

4. Resolve the corrected Conjecture 5.3 for $X = \mathbb{R}$ under strict coordinatewise monotonicity alone, including the proposed automatic continuity conclusion.

5. Construct infinitely many pairwise nonconjugate associative, idempotent, strictly monotone ternary operations on $\mathbb{Q}$, distinguishing them by explicit equations expressing prime divisibility.

Verification. The companion exact verifier checks the universal linear identities, the binary-reduction obstruction, and the midpoint calculation. Existence of the infinite bijection and its order properties are proved above, not inferred from finite experiments. Review is by the authoring agent; this is not independent peer review or a proof-assistant certificate.

References

[1] G. Kiss and G. Somlai, *Associative idempotent nondecreasing functions are reducible*, Semigroup Forum **98** (2019), 140–153. doi:10.1007/s00233-018-9973-y. Conjecture 5.3 and Example 4.10; also arXiv:1707.04341v3. The 16-page postprint deposited at ORBilu 10993/34510 retains the conjecture on page 15.